Project Management

Cost Benefit Analysis, roles, and finance

Roles in A Project

A successful project requires a clearly defined organizational structure to balance business vision, resource management, and technical execution.

Leadership & Governance:

- *Project Sponsor / Owner:* The stakeholder who champions the project, secures funding, and defines the ultimate business objectives.
- *Project Manager (PM):* The individual responsible for balancing project constraints (Time, Cost, Scope), managing risk, and driving execution.

Delivery & Execution:

- *Technical Lead / Architect:* Translates business requirements into a concrete technical design or system engineering blueprint.
- *Business Analyst (BA):* Acts as a bridge between end-users and implementation teams, turning vague stakeholder requests into structured specifications.

Project Management

There are many methods to manage a project; however, one of the core rules of project management is:

> Adopt a methodology and use it consistently.

A project manager's main jobs are *measuring*, *evaluating*, and *correcting*. This is often done through high-level supervision and reporting.

This involves generating deliverables like:

- Monthly financial summaries
- Milestone progress reports
- Phase-gate testing evaluations

The PM's ultimate goal is to ensure the project continues to deliver *business value* and aligns with the original corporate strategy.

IT Financial Management

Financing a project goes beyond the initial software purchase. It requires understanding modern corporate IT budget models.

Budgeting Approaches:

- *CapEx (Capital Expenditure):* Upfront, one-time investments in fixed assets (e.g., buying physical on-premise servers or perpetual software licenses).
- *OpEx (Operating Expenditure):* Ongoing, recurring operational costs (e.g., monthly AWS/Azure cloud consumption bills or SaaS subscription fees).

Total Cost of Ownership (TCO):

- A financial estimate calculated to uncover *hidden lifecycle costs*, including employee training, security patches, system maintenance, and decommissioning fees.

Cost-Benefit Analysis (CBA)

A systematic process used to calculate whether the financial return of an integration architecture *justifies* its implementation costs.

How to Execute a CBA:

1. **Identify Costs:** Log all upfront and recurring costs (software licenses, developer labor, server hardware).
2. **Estimate Benefits:** Quantify financial gains (e.g., hours saved through automation, increased sales capacity, reduced server footprints).
3. **Map the Timeline:** Plot costs against benefits over a multi-year horizon (typically 3 to 5 years).
4. **Clarify Risks:** Factor in potential financial impacts from risks (regulatory fines, market competition, environmental downtime).

Cost-Benefit Analysis (CBA) Metrics

Common Quantitative Methods (with Examples):

- **Return on Investment (ROI):** The percentage efficiency of the investment.
  - *Example:* A $100,000 project that generates $150,000 in value yields a 50% ROI.
- **Payback Period:** The exact time it takes for the project's cumulative benefits to break even with its costs.
  - *Example:* An $80,000 software tool that saves $20,000 a month has a 4-month payback period.
- **Net Present Value (NPV):** Calculating future cash flows in terms of today's inflation-adjusted dollar value.
  - *Example:* Factoring in inflation to prove that saving $10,000 three years from now is worth less than saving $10,000 today.

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Case Study 1: The Cloud SaaS Migration (CapEx vs. OpEx)

A company considers moving from aging internal servers to a managed Cloud CRM (Salesforce).

- *The Costs (TCO):* $50,000/year in cloud subscription fees (OpEx) + $20,000 one-time data migration cost.
- *The Benefits:* Eliminates $30,000/year in physical server maintenance, saves $15,000/year in electricity, and increases sales team revenue by $40,000/year due to better mobile access.
- *The CBA Outcome:* The Year 1 costs ($70,000) are offset by Year 1 benefits ($85,000). The payback period is immediate (under one year), justifying the switch to an OpEx model.

Case Study 2: Process Automation Integration

An HR department wants to integrate an automated payroll system to replace manual data entry.

- *The Costs:* $100,000 to build the custom API integration (CapEx) + $5,000 annual maintenance.
- *The Benefits:* Frees up 3 HR employees to do other work, saving $120,000 annually in labor/overhead costs.
- *The CBA Outcome:* - *ROI:* Highly positive over a 3-year timeline.
    - *Payback Period:* 10 months ($10,000 in monthly labor savings against a $100,000 initial cost).

Approved. The initial sticker shock is high, but the long-term TCO is much lower than the manual baseline.

Case Study 3: ERP Integration

A retail chain attempts to build a custom on-premise Enterprise Resource Planning (ERP) system from scratch instead of buying a proven vendor solution.

- *The Costs:* Initially budgeted at $2,000,000, but inflated to $5,000,000 over 3 years due to poorly defined requirements and constant developer rework.
- *The Benefits:* Projected to save $500,000 annually in supply chain efficiencies, but zero actual benefits were realized because the system could never pass integration testing.
- *The CBA Outcome:* - *ROI:* Deeply negative (Total capital loss of $5,000,000).
  - *Payback Period:* Infinite (the system never launched to generate savings).

Abandoned. The lack of strict requirements, contracts, QA, project management, and a failure to enforce the original project scope turned the integration into a catastrophic sunk cost.

Hertz vs Accenture

https://www.henricodolfing.ch/case-study-8-how-hertz-paid-accenture-32-million-for-a-website-that-never-went-live/